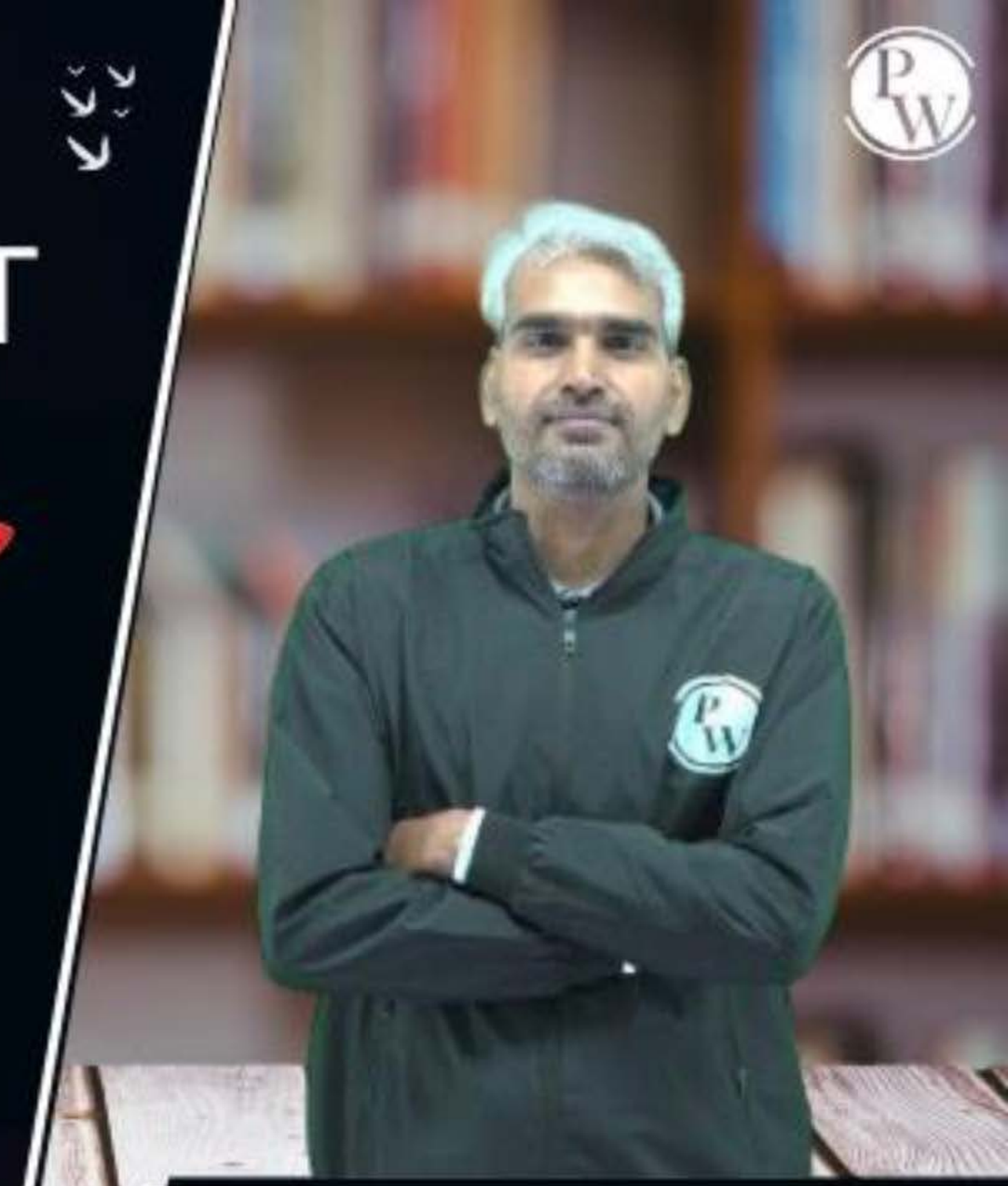
# Computer Science & IT



## Discrete Mathematics

## Set Theory & Algebra

**Lecture No. 13**



By- Vishal Sir

# Recap of Previous Lecture



- Topic: Partial order relation
- Topic: Partially ordered set (POSET)
- Topic: Comparability
- Topic: Total order relation and TOSET

Slide

# Topics to be Covered



- Topic: Least upper bound (LUB)
- Topic: Greatest lower bound (GLB)
- Topic: Minimal and maximal elements in a POSET
- Topic: Minimum (Least) element of POSET
- Topic: Maximum (Greatest) element of POSET
- Topic: Lattice

Slide

# Today's Topic

**Note:** Let $(A, R)$ is a POSET.

and $a, b \in A$

and $aRb$, i.e. $(a, b) \in R$ { i.e. a & b are Comparable w.r.t. POSET $(A, R)$ }

Element 'a' is at lower side of 'b'

Element 'b' is at upper side of 'a'



# Topic : Least Upper Bound / lub / Join / Supremum

- Let $(A,\leq)$ be a POSET, for any two elements $a,b\in A$ if there exists an element $c\in A$ such that,

  $a\leq c$ and $b\leq c$, ⟶ then $c$ is called upper bound of $a$ and $b$,

  $(a,c)\in\leq$ & $(b,c)\in\leq$

  element 'c' is at upper side of both 'a' & 'b'

  And if there exists no element $d\in A$ such that

  $a\leq d$ and $b\leq d$ and $d\leq c$,

  $((a,d)\in\leq$ & $(b,d)\in\leq)$ & $(d,c)\in\leq$ ⟶ then $c$ is called least upper bound of $a$ and $b$

  'd' is the upper bound of a & b (and) 'd' is at lower side of 'c'

Slide



# Topic : Least Upper Bound / lub / Join / Supremum

it can be any partial order relation 'R' on set A

- Let $(A,R)$ be a POSET, for any two elements $a,b \in A$ if there exists an element $c \in A$ such that,

  $aRc$ and $bRc$, → then $c$ is called upper bound of $a$ and $b$,

  $(a,c) \in R$ & $(b,c) \in R$

  element 'c' is at upper side of both 'a' & 'b'

  And if there exists no element $d \in A$ such that

  $aRd$ and $bRd$ and $dRc$,

  $(a,d) \in R$ & $(b,d) \in R$ & $(d,c) \in R$

  'd' is the upper bound of a & b (and) 'd' is at lower side of 'c'

  then $c$ is called least upper bound of $a$ and $b$

Slide



# Topic : Greatest Lower Bound / glb / Meet / Infimum

Let $(A, \leq)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$c \leq a$ and $c \leq b$, → then $c$ is called lower bound of $a$ and $b$,

$(c, a) \in \leq$ & $(c, b) \in \leq$

'c' is at lower side of both 'a' & 'b'

And if there exists no element $d \in A$ such that

$d \leq a$ and $d \leq b$ and $c \leq d$,

$(d, a) \in \leq$ & $(d, b) \in \leq$ & $(c, d) \in \leq$ then $c$ is called greatest lower bound of $a$ and $b$

('d' is the lower bound of a & b) and 'd' is at upper side of c

Slide



# Topic : Greatest Lower Bound / glb / Meet / Infimum

any partial order relation 'R' on set A

Let $(A, R)$ be a POSET, for any two elements $a, b \in A$ if there exists an element $c \in A$ such that,

$c R a$ and $c R b$, ⟶ then $c$ is called lower bound of $a$ and $b$,

$(c,a) \in R$ & $(c,b) \in R$

'c' is at lower side of both 'a' & 'b'

And if there exists no element $d \in A$ such that

$d R a$ and $d R b$ and $c R d$,

$(d,a) \in R$ & $(d,b) \in R$ & $(c,d) \in R$ then $c$ is called greatest lower bound of $a$ and $b$

('d' is the lower bound of a & b) and 'd' is at upper side of c

Slide

* Least upper bounds of two elements a & b can be denoted by $lub(a,b)$ or $a \vee b$

$$lub(a,b) = lub(b,a)$$

Notation for least upper bound

**Note:-** Greatest lower bound of two elements a & b Can be denoted by $glb(a,b)$ or $a \wedge b$

↑ Notation for glb

$$glb(a,b) = glb(b,a)$$

eg: Let $A = \{1, 2, 3, 4\}$, and $(A, \leq)$ is a POSET

then lub(2,3) = ? & glb(2,3) = ?

$2 \nleq 1$ $2 \leq 2$ $2 \leq 3$ $2 \leq 4$

$3 \nleq 1$ $3 \nleq 2$ $3 \leq 3$ $3 \leq 4$

'3' & '4' are at upper side of both "2 & 3"

∴ Both 3 & 4 are upper bounds of '2 & 3'

- $3 \leq 4$, i.e 3 is at lower side of 4
  ∴ '4' can not be the lub(2,3)
- W.r.t. '3', there is no element at the lower side of '3' {apart from '3' itself} which is the upper bound of 2 & 3

∴ lub(2,3) = 3

$1 \leq 2$ $2 \leq 2$ $3 \nleq 2$ $4 \nleq 2$

$1 \leq 3$ $2 \leq 3$ $3 \leq 3$ $4 \nleq 3$

'1' & '2' both are lower bounds of '2 & 3'

But $1 \leq 2$ (2 is at upper side of 1)

∴ '1' can not be the glb(2,3)

- W.r.t. '2' there is no other element at upper side of 2 {apart from '2' itself}, which is lower bound of '2 & 3'

∴ glb(2,3) = 2

**Note:-** If $A$ is any set of real numbers and $(A, \leq)$ is a POSET, then for two elements $a, b \in A$

"less than Equal"

$$lub(a,b) = Max(a,b)$$

$$glb(a,b) = Min(a,b)$$

Let A is a set of all positive integers and $(A, \div)$ is a POSET.

'divides'

for two elements $a, b \in A$

$$lub(a,b) = LCM(a,b)$$

$$glb(a,b) = GCD(a,b)$$

**Note :→**

$\div$

$(x, y) \in \div$

iff x divides y

or

y divided by x

- 'y' is at upper side of x iff 'y' is divided by x

&

- 'x' is at lower side of y. iff x divides y

**Note:-** If $A$ is any set of Positive integers and $(A, \div)$ is a POSET.

- then for two elements $a, b \in A$

$lub(a, b)$ = Nothin can be said
Because we don't know the elements of set A

$glb(a, b)$ = Nothin can be said
Because we don't know the elements of set A

Note:→

If $A \subseteq B$,
then A is at lower side of B
&
B is at upper side of A

Let A is any finite set and P(A) is the power set of set A, then $(P(A), \subseteq)$ is a POSET.

(P(A): Collection of sets)

For two elements (sets) $X, Y \in P(A)$

$lub(X, Y) = X \text{ union } Y$

$glb(X, Y) = X \text{ intersection } Y$

**Note:-** Let $C$ is any Collection of sets. then $(C, \subseteq)$ is a POSET

for two sets $X, Y \in C$

$lub(X, Y)$ = Nothing can be said
Because we don't Know the elements of set $C$

$glb(X, Y)$ = Nothing can be said
Because we don't Know the elements of set $C$

eg: $C = \{\{1\}, \{2\}, \{1,2,3,4,5\}\}$

& $(C, \subseteq)$ is a POSET

$lub(\{1\}, \{2\}) = \{1,2,3,4,5\}$

$glb(\{1\}, \{2\})$ = Does not exist

eg:- let $A = \{1, 2, 3, 5, 24\}$
and $(A, \div)$ is a POSET

$lub(2,3) = 24$

$lub(3,5) =$ Does not exist

$glb(2,3) = 1$

$glb(3,5) = 1$

eg: Let $C = \{ \{1\}, \{2\}, \{1,2,3\}, \{1,2,4\} \}$
and $(C, \subseteq)$ is a POSET.

$lub(\{1\}, \{2\})$

⇓

We can observe both $\{1,2,3\}$ & $\{1,2,4\}$ are upper bounds of "$\{1\}$ & $\{2\}$"

Some author defines, both $\{1,2,3\}$ & $\{1,2,4\}$ as lub of "$\{1\}$ & $\{2\}$"

"and"

W.r.t. some authors $lub(\{1\}, \{2\})$ does not exist ⇐ they argue that "least" must be unique

Note :→

In a POSET lub and/or glb need not exist for Every Pair of elements



# Topic : Minimal elements in a POSET

In a POSET there may be more than '1' Minimal elements

R — It can be any Partial order rel$^n$ R on set A.

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a minimal element of POSET$(A, \leq)$ if there exists no element $y \in A$ other than $x$ itself, such that $y \leq x$ . i.e. $(y,x) \in \leq$ $(A,R)$

$y \, R \, x$ i.e. $(y,x) \in R$

Slide



# Topic : Minimal elements in a POSET

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a minimal element of POSET$(A, \leq)$ if there exists no element $y \in A$ other than $x$ itself, such that $y \leq x$ .

i.e., if there is no element 'y' other than 'x' itself which is at the lower side of x, then 'x' is a minimal element of the POSET.

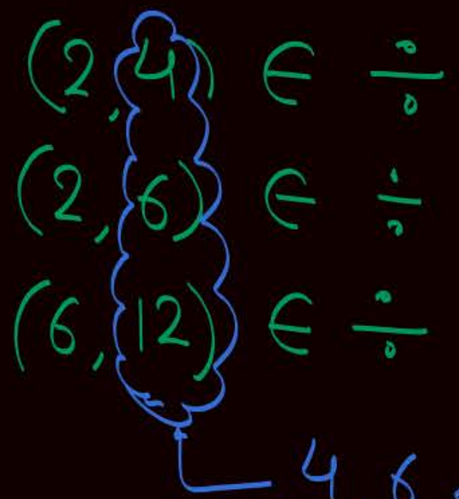
Slide

eg:- $A = \{2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

find all minimal elements in the POSET.

'2' & '3' are the minimal elements of POSET





# Topic : Maximal elements in a POSET

(A, R) it can be any partial order relation R

"More than one also possible"

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a maximal element of POSET$(A, \leq)$ if there exists no element $y \in A$ other than $x$ itself, such that $x \leq y$ . i.e $(x,y) \in \leq$ (A, R)

↳ $x R y$ i.e $(x,y) \in R$

Slide



# Topic : Maximal elements in a POSET

Let $(A, \leq)$ be a POSET

An element $x \in A$ is called a maximal element of POSET$(A, \leq)$ if there exists no element $y \in A$ other than $x$ itself, such that $x \leq y$ .

If there is no element 'y', other than 'x' itself which is at the upper side of x, then x is the maximal element of POSET.

Slide

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find all maximal elements in the POSET.

4, 6 & 15 are maximal elements of POSET

eg:- $A = \{2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find all maximal and minimal elements of the POSET

Maximal element is '12'

and

Minimal elements are "2 & 3"

eg:- $A = \{1, 2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find all minimal and maximal elements of POSET.

Minimal is '1'

Maximal is '12'

Imp:

Q:. Let $A = \{1, 2, 3, 4, 6\}$

and $(A, \geq)$ is a POSET.

"greater than equal"

$(2 \geq 1)$

then '1' is at upper side & '2' is at lower side

$lub(2,3) = 3 = 2$

$glb(2,3) = 2 = 3$



# Topic : Minimum element of a POSET

(A, R) (least)

Let $(A, \leq)$ be a POSET

An element $m \in A$ is called a minimum element of POSET$(A, \leq)$ (A, R)

if and only if

$m \leq a, \forall a \in A$ . i.e $(m, a) \in \leq \ \forall a \in A$

i.e $m \, R \, a, \forall a \in A$ , i.e $(m, a) \in R, \forall a \in A$

'm' is at lower side of all elements of set A

Slide

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find minimum element of POSET.

No element of set "divides" all elements of set A.

∴ Minimum element does not exist in this POSET

In the given POSET '2 & 3' are minimal elements.

Note:- If there exists two or more minimal elements in a POSET, then minimum element can never exist in that POSET.

eg:- $A = \{1, 2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find minimum element of POSET.

'1' is the only "minimal" element of POSET

&

'1' divides all the elements of the set A,

∴ '1' is also the "minimum" element of POSET

Note:- If there exists a unique minimal element in the POSET, then that element will also be the minimum element of POSET.

Note: Minimum element if exists, then it is always unique.



# Topic : Maximum element of a POSET

(Greatest)

(A, R)

Let $(A, \leq)$ be a POSET

(A, R)

An element $n \in A$ is called a maximum element of POSET$(A, \leq)$

if and only if

$a \leq n, \forall a \in A$ i.e $(a, n) \in \leq, \forall a \in A$

$a R n, \forall a \in A$, i.e. $(a, n) \in R, \forall a \in A$

Slide

eg:- $A = \{2, 3, 4, 6, 15\}$

and $(A, \div)$ is a POSET.

Find Maximum element of POSET.

* There is no element in set A which is "divided by" all elements of set A.

  ∴ Maximum element does not exist in the POSET.

- In the given POSET '4', '6' & '15' are maximal elements

Note:- If there exists two or more maximal elements in a POSET, then maximum element can never exist in that POSET.

eg:- $A = \{1, 2, 3, 4, 6, 12\}$

and $(A, \div)$ is a POSET.

Find maximum element of POSET.

'12' is the only maximal element of POSET

'and'

'12' is divided by all the elements of set A.

∴ '12' is also the maximum element of POSET.

Note:- If there exists a unique maximal element in the POSET, then that element will also be the maximum element of POSET.

Note: Maximum element if exists, then it is always unique.

**Note:-** In a POSET, minimum and/or maximum element need not exist, but if exists then they will be unique.

Q:. Is it necessary that minimal and maximal elements exist in a POSET

↳ Ans = NO.

"Explaination" => "Next Slide"



## Topic : NOTE

infinite set

- Let $Z$ is the set of all integers, then $(Z, \leq)$ is a POSET.

infinite set — $Z$ ; Rel$^n$ = "less than or equal" — $\leq$

- No minimal element exist in POSET
  Hence No minimum element as well

- No maximal element exist in POSET
  Hence No maximum element as well

Slide



**Topic : NOTE**

- Let N is the set of all natural numbers, then $(N, \leq)$ is a POSET.

  "let it starts from 1"

  - '1' is the only minimal element of POSET.
    And hence '1' is also the minimum element of POSET.

  - But No maximal or maximum element exist in POSET.

Slide

* If there is no minimal element in the POSET and/or no maximal element in the POSET, then set is guaranteed to be an infinite set.

* But if set is an infinite set then minimal and/or maximal elements may or may not exist



**Topic : NOTE**

eg: Let $A = \{ x \mid x \in R \text{ and } 0 \leq x \leq 1 \}$

A is an infinite set of real nos

And, $(A, \leq)$ is a POSET

Here '0' is the minimal as well as minimum element of POSET

&

'1' is the maximal as well as maximum element of POSET.

Slide

**Note :-** ① If set is a finite set, then in a POSET w.r.t. that set.

there will be at least one minimal element

and there will be at least one maximal element

② In a POSET {may or may not be finite}

there can be at most one minimum element

And there can be at most one maximum element



# 2 mins Summary

- **Topic** Least upper bound (LUB)
- **Topic** Greatest lower bound (GLB)
- **Topic** Minimal and maximal elements in a POSET
- **Topic** Minimum (Least) element of POSET
- **Topic** Maximum (Greatest) element of POSET
- **Topic** Lattice

# THANK - YOU